



COMP0016: Systems Engineering

Final Prototype Deliverables and

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COMP0016 Assessment Components

Component Title	Weight	Submission Deadline
Coursework (HCI)(group)	5%	Nov 2025
Presentation (pitch)(group)	5%	Jan 2026
Website Report (Group with Individual Contribution)	40%	16:00 Friday 27 Mar 2026
System Artefacts (Group with Individual Contribution)	50%	16:00 Friday 27 Mar 2026

Example Websites

- <https://students.cs.ucl.ac.uk/2024/group1/>
- <https://students.cs.ucl.ac.uk/2024/group4/>
- <https://students.cs.ucl.ac.uk/2024/group14/>
- <https://students.cs.ucl.ac.uk/2024/group15/>
- <https://students.cs.ucl.ac.uk/2024/group28/>
- <https://students.cs.ucl.ac.uk/2023/group1/>
- <https://students.cs.ucl.ac.uk/2023/group23/>
- <https://students.cs.ucl.ac.uk/2020/group8/>
- <https://students.cs.ucl.ac.uk/2019/group4/>
- <https://students.cs.ucl.ac.uk/2018/group24/>

Website Structure

- Home
- Requirements
- Research
- Algorithms (if applicable)
- UI Design (if applicable)
- System Design
- Implementation
- Testing
- Evaluation
- Appendices
 - User manual and deployment manual
 - GDPR and Privacy of Data
 - Development Blog – External Link on a public site (**this should be a link to a public site, not a term 2**)
 - Monthly Video

Website Submission

- You can use **any tools or frameworks** to develop your re website should be a static website that **only includes the**
- **The examiners should not be required to install anything file to navigate all the pages.**
- **Please ensure all the links work** – especially for the ind **your pages.** We won't open the html files that are not link

Home

- Project title
- An abstract should include three paragraphs and the following
 - Problem statement
 - Your solution
 - Achievement and impact
- A 8 minutes video to introduce the project and go through the project
- The development team
 - (including the team photos where possible) name, email, phone number of each member. The roles include client liaison, UI designer, front-end developer, editor, tester.
 - You are welcome to include interesting background and LinkedIn profiles if you wish.
- Project management
 - Gantt chart (from October 20 2025 to March 27 2026)

Issues on the Home Page

- The abstract is too brief and requires additional detail.
- The project background and motivation are not included.
- The video is missing.
- The provided video is too short and does not effectively showcase
- The resolution of the live demo is too low, making it hard to see the
- The narration is difficult to hear due to loud background music or a
- The video resolution is too low, making it hard to view the content
- The Gantt chart is displayed too small, making the text difficult to r

Requirements

- Partner Introduction and Project background
- Project Goals
- Requirement Gathering
 - How did you collect the requirements?
 - Did you design any survey? How did you analyse the survey?
- Personas
 - Typical users of your project
- Use Cases (if applicable)
 - Use case diagram
 - List of use cases
- MoSCoW Requirement List (Functional and Non-functional)
 - A table for function requirements
 - Another table for non-functional requirements: https://en.wikipedia.org/wiki/MoSCoW_functional_requirement. For example, performance, security, maintainability, extensibility, etc.

Issues on the Requirements P

- The use case diagram is missing.
- The requirements are described too briefly, typically consisting of one sentence. They should be expanded to provide sufficient detail and clarity.
- The submitted use case diagram does not adhere to the proper format for a use case diagram.

MoSCoW requirement list

- **Functional** requirements

ID	Requirements
1	
2	
3	

- **Non-functional** requirements

ID	Requirements
1	
2	
3	

Requirement Example

2.6.1 Functional Requirements: General

ID	Requirement
FR-G1	The app shall allow unregistered users to register an account.
FR-G2	The app shall allow registered users to log in to their accounts, and show they are logged in.
FR-G3	The app shall display a 'Home' page and browsing pages 'Projects', 'News' and 'Events' that are accessible without login.
FR-G4	The app shall allow any user to watch the project video for a given project.
FR-G5	The app shall allow any user to view a news article for a given news item.
FR-G6	The app shall allow any user to view event details for a given event item.
FR-G7	The app shall display an 'About' page that introduces the web application, providing context and background.
FR-G8	The app shall allow the user to search and filter on browsing pages.
FR-G9	The app shall allow registered users to edit their account details.
FR-G10	The app shall allow registered users to upload and change their own profile picture.
FR-G11	The app shall provide users with the option to reset their password if forgotten.
FR-G12	The 'Home' page or landing page shall showcase recent projects, published news and live events.
FR-G13	The app may send email notifications for events such as welcome message and password resets as part of engaging with users.

Requirement Example Continue

Functional Requirements - Admin

ID	Functional Requirements - Admin
FR-A1	The app shall display all campaigns on the app, including those created by the Vendors, on the main page.
FR-A2	The app shall allow the Admin to view the details and data analytics of all marketing campaigns on the app.
FR-A3	The app shall allow the Admin to suspend a campaign upon requests from a Vendor.
FR-A4	The app shall allow the Admin to view a data analytics overview for all marketing campaigns including the total number of views, the total number of clicks, the number of conversions over a period, the number of clicks over a period, budget remaining, and campaign time.
FR-A5	The app shall allow the Admin to edit account privileges.
FR-A6	The app shall allow the Admin to edit all marketing campaigns.
FR-A7	The app shall allow the Admin to send and receive messages from Vendors.
FR-A8	The app shall notify the Admin of new messages from Vendors.
FR-A9	The app shall allow the Admin to deactivate/activate Vendor accounts.
FR-A10	The app shall allow the Admin to generate invitation codes required to register new Vendor accounts.
FR-A11	The app shall allow an Admin to create new Admin accounts.
FR-A12	The app shall notify Admins when a message sent to a Vendor is not seen for a specified amount of time.

Research

- Related Projects Review
 - Review at least 1 existing similar project if applicable
 - For each of the similar projects, please list the project name, I will be using from this existing application.
- Technology Review
 - Please compare the possible solutions, describe what you choose
 - Please compare the possible devices (if applicable), describe what you choose
 - Please compare the possible algorithms (if applicable), describe what you choose
 - Please compare the possible programming languages, frameworks, and tools you choose, and explain why.
- A summary of your technical decisions
- References
 - IEEE style: reference definition needs the number
"Several recent studies [1, 4, 15, 22] have suggested that. . . ."
[1] L. Bass, P. Clements, and R. Kazman, *Software Architecture in Practice*, Reading, MA: Addison Wesley, 2003. [E-book] Available: Safari e-book
 - IEEE citation reference definition
 - <https://iee-dataport.org/sites/default/files/analysis/27/IEEE>

Common Issues on the Research

- The Review section is quite brief and lack details and should be expanded.
- The organization of the section needs improvement, as the discussion is not clear.
- A comprehensive summary of the technologies employed has not been provided.
- References are absent.
- References are not cited within the body of the text using IEEE style.

Algorithms (if applicable)

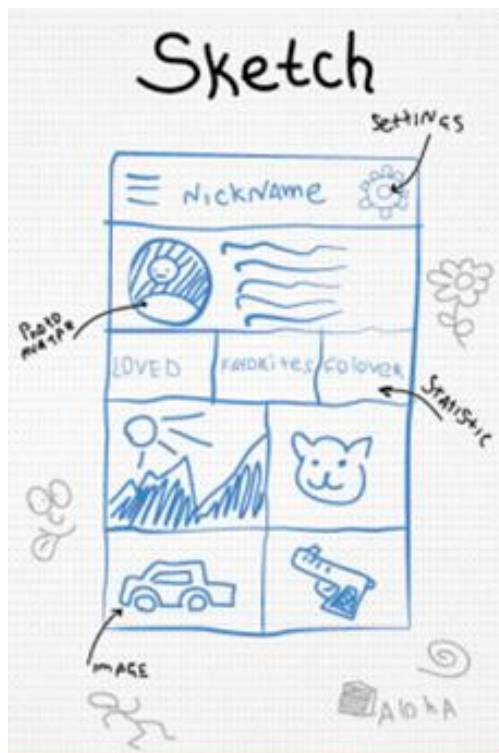
- Models
 - Describe the key idea of your chosen algorithms
- Data
 - Dataset
 - Data Preprocess (if applicable)
 - Training and testing sets
- Experiments
 - Experiment Design
 - Performance Evaluation Method (e.g. % accuracy)
 - Experiment results that should be presented by quantified values
 - Investigation of the optional Hyperparameters (if applicable)
 - Use plots or tables to show the performance comparison results
- Discussions
 - Why the algorithm fails for some test examples
 - Suggestions to improve the performance
- Conclusion
- References

Common Issues on the Algor

- The performance evaluation metric is not explained clearly.
- A quantitative evaluation has not been provided.
- No references are included.
- References are not cited within the text according to IEEE formatting.

User Interface Design (if applicable)

- Design Principles (e.g. simplicity, consistency, visibility)
- Hand-drawn sketches
- Online interactive wireframe URL (High Fidelity wireframes)



Wireframe



System Design

- System architecture diagram and a brief description of each component
- Site Map (if applicable)
- Sequence Diagrams (if applicable)
- Design Patterns (if applicable)
- Class Diagrams (if applicable)
- Data storage (if applicable)
 - Please provide the data schema (e.g. ER diagram) if applicable
- Packages and APIs defined (if applicable)

Common Issues on the System

- Several components are missing from the system architecture diagram
- The ER diagram is not presented in the correct format and contains errors

Implementation (Very Important)

- Please describe how you implement the key features
 - It is very hard for the other people to understand your code. You need to provide the necessary explanation for other people to understand how you implement the key features. Describe the frameworks or plugins or libraries you used for a particular feature.
 - Please describe one key feature in a subsection. We can use an example, you can describe the implementation details for a feature like authentication, database connection, data table, trend generation, notification, etc.
 - If necessary, sequence or other diagram and short code snippets with explanation.

Common Issues on the Imple

- Some images fail to load.
- The system architecture diagram needs improvement. Some core

Testing

- Testing Strategy
- Unit and Integration testing
- Compatibility testing (if applicable)
- Responsive design testing (if applicable)
- Performance/stress testing (if applicable)
- Prompt Testing (if applicable)
- User acceptance testing
 - Simulated testers
 - Test cases
 - Feedback from testers and project partners
- For each test, you should explain why you do this test, when to conduct the test, what results you get, and your analysis of the results

Common Issues on the Testin

- Details on how the tests were conducted are missing.
- Integration testing should be automated rather than relying solely on manual testing.
- Feedback from the project partner should be provided in the user stories.

Evaluation

- Summary of achievements
 - An achievement table to list the MoSCoW functionalities and their contributors
 - A list of known bugs (if applicable)
 - Individual contribution distribution table
- Critical evaluation of the project
 - User interface / user experience (if applicable)
 - Functionality
 - Stability
 - Efficiency
 - Compatibility
 - Maintainability
 - Project management
- Future work
 - How could the project be extended if you have more time?

Common Issues on the Evaluation

- Some requirements were not implemented but has been incorrectly
- Separate individual contribution tables for the system development provided. Currently, only a combined table is included.
- The current "Future Work" section is quite brief. It would be better on potential enhancements and next steps.

Example of Achievement Table and Known bug list

- Achievement table

ID	Requirements	Priorities
1		Musical
2		Musical
3		Short
4		Could
Key Functionalities (must have and should have)		
Optional Functionalities (could have)		

- Known bug list

ID	Bug Description
1	
2	
3	

Individual Contribution Table For System

Work packages	John	
Research and Experiments	40%	
UI Design (if applicable)	20%	
Coding	40%	
Testing	20%	
Overall contribution	30%	

Individual Contribution Table For Website

Work packages	John	
Website Template and Setup	0%	
Home	33%	
Video	33%	
Requirement	34%	
Research	50%	
Algorithm (if applicable)	20%	
UI Design (if applicable)	20%	
System Design	20%	
Implementation	30%	
Testing	0	
Evaluation and Future Work	60%	
User and Deployment Manuals	20%	
Legal Issues	100%	
Blog and Monthly Video	33%	
Overall contribution	30%	

Appendices

- User manual
 - Teach users to use your application
 - It would be very helpful for the readers if some screenshots are provided
 - If you need to deliver a website, please remember to provide a default username, and password to login your website if authentication is required
 - If there are more than one type of user (e.g. admin and normal user), please provide instructions for each type of user.
- Deployment manual (**very important**)
 - This should provide a step by step guide to show your project how to be deployed so that they get your source code.
 - If needed for some difficult parts, please provide some screenshots
- Legal issues and processes
 - Notes have been uploaded already to Moodle for you make use of
- Development Blog
- Monthly Video

Common Issues on the Appeal

- If applied, the URL of the deployed online application is missing and
- The test accounts are not provided.
- Screenshots should be added to support the user manual for clarification.

Legal Issues and Processes

- The notes for this section will be available in “Legal and GDPR” section on Moodle.
- There is an extensive deck of notes for you to read.
- Your website should include GDPR and privacy of course. **document this on your project website under Appendix**
- Include a statement on the source code license (choose one first) and GDPR of data governance.

Development Blog

- This is an open and public summary of your team's progress, your websites' key assets, progress and experimental design. You can use your own preferred blog, or any of the UCL blogging tools.
- The Project Partners have asked to keep track of your progress. Ideas for the blog would be to include Interesting research findings, materials, sketches (including from your HCI components), code snippets especially what you have found and customised solutions. Of course your own diagrams and models should be posted.
- Check it with your Project Partners. You can post your progress on social media with their emailed permission. Each student should contribute to this.
- As they are a diary, you should be aiming to update the blog weekly (do not submit it all at the end!).

Monthly Video

- Each month you should ideally be preparing a 1-2 min interesting findings and progress made on your experience.
- Use the PowerPoint PPTX template found in the “Project” section as the start for your video (3-5 seconds).
- Please remember to set the OneDrive sharing setting to ‘with the link’. If you need a date for your diaries to remember each month.
- Please submit the OneDrive link on the form provided on the website. UCL will put approved videos onto their YouTube channel for make use of.

Report Website Marking Criteria

- **Home:** Is the abstract explained well? How good is the visualisation?
- **Requirements:** Is the project background and goals explained? Are the requirements captured and described? Are personal and MoSCoW requirements clear and specific?
- **Research:** How well the related project reviewed? How well the software, API, libraries, algorithms? Are references in place?
- **Algorithms** (if applicable): Is the model, experiments, results explained well? Did you build up a quantified scheme to evaluate the results?
- **UI Design** (if applicable): Is the UI design principles, sketches, and wireframes explained?
- **System Design:** Has the system architecture design been explained with a diagram and suitable diagrams.

Report Website Marking Criteria

- **Implementation:** How well is the implementation of key features?
- **Testing:** Is there a good testing strategy and a thorough test plan?
- **Evaluation:** Is there a good evaluation of the end results? Are the results relevant, and the conclusions justified? Is the future plan viable and concrete?
- **Appendix:** Are the user and deployment manuals clear? Are screenshots and month videos included?
- **Format:** Does the website have a good format?
- **Clarity:** Is the content well written and readable? How are the results reported? Does it communicate effectively?

Mark Range for Report Website

Mark Range	Description	Comments
90 - 100	Exceptional	Publishable quality. Close to
80 - 89	Outstanding	Could lead on to publishable
70 - 79	Distinction	Very well written with a clear
60 - 69	Good (merit)	Clear project write-up with lo
50 - 59	Satisfactory (pass)	Adequate project write-up, la containing irrelevant materia
40 - 49	Weak (BSc and MEng Pass)	Write-up is somewhat incohe omissions, or irrelevant mate
30-39	Inadequate(Fail)	Documentation is poor, unst
0 - 29	Unacceptable fail	Documentation is substantial incomprehensible or wrong.

Hosting the Project Website

- Hosting Space

- Each team is assigned a 500 MB space for hosting website.
- Your team's specific directory will follow the format **/cs/student/www/2025/group<id>/**, where <id> is your team ID.
- Others can visit your project website using a URL to **http://students.cs.ucl.ac.uk/2025/group<id>**